



Post Office Box 2000, Decatur, Alabama 35609-2000

November 25, 2024

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 2
Renewed Facility Operating License No. DPR-52
NRC Docket No. 50-260

Subject: Licensee Event Report 50-260/2024-002-01 – High Pressure Coolant Injection Inoperable and Isolated due to Rupture Disc Failure

- References:**
1. Letter from NRC to TVA, "Browns Ferry Nuclear Plant – NRC Inspection Report 05000260/2024090; and Apparent Violation" dated September 17, 2024. (ML24255A027)
 2. Letter from TVA to NRC, "Licensee Event Report 50-260/2024-002-00 – High Pressure Coolant Injection Inoperable due to Rupture Disc Failure and Resulting System Isolation," dated May 20, 2024 (ML24141A246)

The enclosed Licensee Event Report provides details of an isolation of the High Pressure Coolant Injection (HPCI) system on Browns Ferry Nuclear Plant, Unit 2. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(v)(D), as any event or condition which could have prevented the fulfillment of the safety function of a system that is needed to mitigate the consequences of an accident; and 10 CFR 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in black ink, appearing to read "DLAK", is written over a horizontal line.

Daniel A. Komm
Site Vice President

U.S. Nuclear Regulatory Commission
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November 25, 2024

Enclosure: Licensee Event Report 50-260/2024-002-01 – High Pressure Coolant Injection
Inoperable due to Rupture Disc Failure and Resulting System Isolation.

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 2**

Licensee Event Report 50-260/2024-002-01

HPCI Inoperable due to Rupture Disc Failure and Resulting System Isolation.

See Enclosed

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027				
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.				
1. Facility Name Browns Ferry Nuclear Plant, Unit 2				☒ 050 ☐ 052		52. Docket Number 00260				
3. Page 1 OF 8										
4. Title High Pressure Coolant Injection Inoperable due to Rupture Disc Failure and Resulting System Isolation										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
03	19	2024	2024	- 002	- 01	11	25	2024	N/A	05000XXX
									Facility Name	Docket Number
									N/A	05000XXX
9. Operating Mode 1				10. Power Level 100						
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)		☐ 21.21(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)				
☐ OTHER (Specify here, in abstract, or NRC 366A).										
12. Licensee Contact for this LER										
Licensee Contact Baruch Calkin, Licensing Engineer								Phone Number (Include area code) 256-278-1031		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS	
D	73	RPD	F103	Y	N/A	N/A	N/A	N/A	N/A	
4. Supplemental Report Expected)					15. Expected Submission Date			Month	Day	Year
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date								N/A	N/A	N/A
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)										
On March 19, 2024, at 1030 Central Daylight Time (CDT), during performance of 2-SR-3.5.1.7, High Pressure Coolant Injection (HPCI) Flowrate at Rated Reactor Pressure, after operating the Unit 2 HPCI system for approximately 37 minutes at rated pressure and flow, BFN Unit 2 received indications of high rupture disc pressure, HPCI Turbine trip, and HPCI isolation. Operations personnel verified HPCI isolated as expected, declared HPCI inoperable, verified BFN Unit 2 Reactor Core Isolation Cooling (RCIC) operability, and entered 2-AOI-64-2B for isolation. On March 19, 2024, at 1719 CDT, Event Notification 57036 was made to the NRC. BFN Unit 2 remained in Technical Specification (TS) Limiting Condition for Operation (LCO) 3.5.1 until the rupture disc was replaced and the HPCI system was declared operable at 1051 CDT on March 25, 2024. The root cause of this event was that the operating ratio of the rupture disc was not used to develop effective operating limits to protect the integrity of the disc. The corrective actions taken to prevent recurrence will be to create or revise a design output document to determine the operating pressure limit for the HPCI and RCIC System turbine exhaust line rupture discs, to revise affected drawings and procedures to specify the operating pressure limit for the rupture discs, and to revise installation and maintenance instructions to include all pertinent operating limits for operating rupture discs on HPCI systems.										

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 2	☒ 050	2. DOCKET NUMBER 00260	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 002	REV NO. - 01

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN) Unit 2 was in Mode 1 at approximately 100 percent power.

II. Description of Event**A. Event Summary**

On March 19, 2024, at 1030 Central Daylight Time (CDT), during performance of 2-SR-3.5.1.7, High Pressure Coolant Injection (HPCI) [BJ] Flowrate at Rated Reactor Pressure, after operating the Unit 2 HPCI system for approximately 37 minutes at rated pressure and flow, BFN Unit 2 received indications of high rupture disc [RPD] pressure, HPCI Turbine [TRB] trip, and HPCI isolation. Operations personnel verified HPCI isolated as expected, declared HPCI inoperable, verified BFN Unit 2 Reactor Core Isolation Cooling (RCIC) [BN] operability, and entered 2-AOI-64-2B for isolation. On March 19, 2024, at 1719 CDT, Event Notification 57036 was made to the NRC. BFN Unit 2 remained in Technical Specification (TS) Limiting Condition for Operation (LCO) 3.5.1 until the rupture disc was replaced and the HPCI system was declared operable at 1051 CDT on March 25, 2024.

The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(v)(D), as any event or condition which could have prevented the fulfillment of the safety function of a system that is needed to mitigate the consequences of an accident; and 10 CFR 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

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NARRATIVE**C. Dates and approximate times of occurrences**

<u>DATE AND APPROXIMATE TIMES</u>	<u>OCCURRENCE</u>
December 14, 2021, 0200 CST	Unit 2 HPCI System declared inoperable for an online maintenance outage.
December 14, 2021, 2227 CST	Maintenance personnel replaced Unit 2 HPCI System Inner and Outer Rupture Discs (BFN-2-RPD-073-0729 & 0730) per PM 500121473.
December 15, 2021, 1336 CST	HPCI Steam Supply valve, 2-FCV-73-16, was placed in the open position. While open, HPCI exhaust pressure rose to a maximum of 161.4 psig.
December 19, 2021, 0300 CST	Unit 2 HPCI System declared operable following successful PMT.
March 19, 2024, 0900 CDT	BFN Unit 2 HPCI declared inoperable for scheduled performance of 2-SR-3.5.1.7, HPCI Main and Booster Pump [P] Set Developed Head and Flow Rate Test at Rated Reactor Pressure. Unit 2 entered TS LCO 3.5.1 Condition C.
March 19, 2024, 1030 CDT	After operating the Unit 2 HPCI system for approximately 37 minutes at rated pressure and flow, BFN Unit 2 received indications of high rupture disc pressure, HPCI Turbine trip, and HPCI isolation. Operations personnel verified HPCI isolated as expected and declared HPCI inoperable.
March 19, 2024, 1719 CDT	EN 57036 made to the NRC.
March 20, 2024, 1151 CDT	Work completed to replace rupture discs 2-RPD-073-0729 and 2-RPD-073-0730 under Work Order (WO) 124373923.
March 25, 2024, 1051 CDT	BFN Unit 2 HPCI declared operable, exited TS LCO 3.5.1 Condition C.
August 22, 2024	It was recognized that the pressure excursion which had occurred in December 2021 most likely resulted in damage to the rupture disc and rendered the Unit 2 HPCI system inoperable.

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NARRATIVE**D. Manufacturer and model number of each component that failed during the event**

The failed disc was a Fike Metal Products 16-inch Rupture Disc, part number 16-CPV-C.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

Relays 73-10A, 73-14A, 73-27A, and 73- 28A picked up at the time of the Unit 2 HPCI Isolation, indicating that pressure switches [PS] had tripped in response to the failure of 2-RPD-073-0729, the Unit 2 HPCI inner rupture disc. Further troubleshooting confirmed that the inner rupture disc had failed and steam was leaking through the disc.

G. The failure mode, mechanism, and effect of each failed component

The inner rupture disc failure was due to exceeding its recommended operational ratio and the lower tolerance of the burst pressure, resulting in the deformation of the rupture membrane and the eventual failure of the rupture disc during the performance of HPCI flowrate testing.

H. Operator actions

There were no operator actions associated with this event.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

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NARRATIVE**III. Cause of the event**

The direct cause of the event was leakage through the 2-RPD-073-0729 inner rupture disc.

A. Cause of each component or system failure or personnel error

The root cause of this event was that the operating ratio of the rupture disc was not used to develop effective operating limits to protect the integrity of the disc.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root cause was identified.

IV. Analysis of the event

The HPCI system is provided to assure that the reactor is adequately cooled to limit fuel cladding temperature in the event of a small break in the nuclear system and loss of coolant which does not result in rapid depressurization of the reactor vessel. The HPCI system permits the nuclear plant to be shut down, while maintaining sufficient reactor vessel water inventory until the reactor vessel is depressurized. The HPCI system continues to operate until the reactor vessel pressure is below the pressure at which Low Pressure Coolant Injection [BO] operation or Core Spray System [BG] operation maintains core cooling.

During the March 2024 event, turbine exhaust header pressure remained considerably less than the Operating Ratio Pressure Limit of 115 psig. HPCI flowrate surveillances were reviewed during the timeframe of December 2021 to December 2023 and the maximum turbine exhaust header pressure was approximately 48 psig. All surveillances from December 2021 – December 2023 were satisfactory and met the acceptance criteria as required by 2-SR-3.5.1.7, and no other pressure transients were recorded that could have challenged the integrity of an undamaged disc. However, the design basis maximum HPCI Exhaust pressure during an accident is 53.3 psig, greater than the tested pressure used when determining that the HPCI system was operable. If an accident were to have occurred from December 15, 2021, to March 19, 2024, Operability of the HPCI System to perform under accident conditions could not be assured.

BFN, Unit 2 TS Limiting Condition for Operation (LCO) 3.5.1, ECCS - Operating, requires that each ECCS injection/spray subsystem and the Automatic Depressurization System (ADS) function of six safety/relief valves shall be OPERABLE in Mode 1, and in Modes 2 and 3 except

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high pressure coolant injection (HPCI) and ADS valves are not required to be OPERABLE with reactor steam dome pressure less than or equal to 150 psig. TS LCO 3.5.1 Condition C.2 requires, when the HPCI system is inoperable, that the system be restored to Operable status within 14 days or in accordance with the Risk Informed Completion Time Program. A Past Operability Evaluation (POE) determined that this condition is likely to have existed since December 15, 2021, which exceeded the 3.5.1.C.2 required completion time.

Additionally, BFN, Unit 2 TS LCO 3.5.3, RCIC System, requires that the RCIC System shall be OPERABLE in Mode 1, and in Modes 2 and 3 with reactor steam dome pressure greater than 150 psig. TS LCO 3.5.3.A.1 requires that, when the RCIC system is inoperable, that the operability of HPCI be immediately verified by administrative means. On several occasions between December 15, 2021, and March 19, 2024, Operations personnel did not identify that HPCI was inoperable when performing an administrative verification of HPCI operability. In each of these instances, RCIC was inoperable for less than twelve hours.

Finally, TS LCO 3.0.4 states that when an LCO is not met, entry into a Mode or other specified condition in the Applicability shall only be made when the associated actions to be entered permit continued operation in the Mode or other specified condition in the Applicability for an unlimited period of time. On March 21, 2023, at 1520 CDT, and also on March 22, 2023, at 0417 CDT, BFN, Unit 2 entered a TS 3.5.1 Applicable Mode when TS 3.5.1 Required Actions were not met.

V. Assessment of Safety Consequences

This event resulted in unplanned inoperability and unavailability of the single train of the BFN, Unit 2, HPCI system to perform its safety function for mitigation of the consequences of an accident. In the event of an emergency, the RCIC system remained operable, and all other ECCS and ADS valves were available during this event to facilitate core cooling except during periods of planned inoperability for maintenance and testing. Based on the above, TVA has concluded that sufficient systems were available to provide the required safety functions needed to protect the health and safety of the public.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

In the event of an emergency, the RCIC system remained operable, and all other ECCS and ADS valves were available during this event to facilitate core cooling except during periods of planned inoperability for maintenance and testing.

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B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shut down.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

The BFN, Unit 2 HPCI system was determined to be inoperable from the time the rupture disc experienced a pressure excursion on December 15, 2021, until the successful completion of the HPCI surveillance on March 25, 2024, approximately two years and one hundred and two days.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's Corrective Action Program (CAP) under Condition Report (CR) 1917980.

A. Immediate Corrective Actions

HPCI rupture discs 2-RPD-073-0729 and 2-RPD-073-0730 were replaced.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

BFN plans to implement the following Corrective Actions to Prevent Recurrence (CAPR):

1. Create or revise a design output document to determine the operating pressure limit for the HPCI and RCIC System turbine exhaust line rupture discs.
2. Revise the following drawings to update the NOTES section to specify the operating pressure limit for the turbine exhaust line rupture discs:
 - a. 1-, 2-, 3-47E812-1, Flow Diagram High Pressure Coolant Injection System
 - b. 1-, 2-, 3-47E610-73-1 Mechanical Control Diagram HPCI System
 - c. 1-, 2-, 3-47E813-1, Flow Diagram Reactor Core Isolation Cooling System
 - d. 1-, 2-, 3-47E610-71-1 Mechanical Control Diagram RCIC System



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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3. Revise the following procedures to specify the operating pressure limit for the turbine exhaust line rupture discs:
 - a. 1-, 2-, 3-OI-71, Reactor Core Isolation Cooling System [Note: Changes needed for Section 3.0 Precautions and Limitations, and Section 8.4. RCIC Turbine Trip]
 - b. 1-, 2-, 3-OI-73, High Pressure Coolant Injection System [Note: Changes needed for Section 3.0 Precautions and Limitations, and Section 8.3 HPCI Turbine Trip]
 - c. 1-, 2-, 3-ARP-9-3B, Alarm Response Procedure [Note: Changes needed for Window 28 alarm response. (RCIC)]
 - d. 1-, 2-, 3-ARP-9-3F, Alarm Response Procedure [Note: Changes needed for Window 25 alarm response. (HPCI)]
4. Revise Vendor Manual BFN-VTD-F103-0050, Installation and Maintenance Instructions for Fike Rupture Disc Assemblies, to include all pertinent operating limits for operating rupture discs on HPCI systems.

Additionally, BFN plans to implement the following Corrective Actions to address contributing causes:

1. Revise Vendor Manual BFN-VTD-F103-0050, Installation and Maintenance Instructions for Fike Rupture Disc Assemblies, to include all pertinent operating limits for operating rupture discs on RCIC systems.
2. Revise ODM 4.7, Operation's Work Control Interface, Attachment 2, to warn against the possibility of over pressurizing low pressure system components during online maintenance, such as when closing the HPCI/RCIC steam exhaust isolation valve with potential valve leak by.

VII. Previous Similar Events at the Same Site

A review of the BFN CAP and Licensee Event Reports (LERs) for Units 1, 2, and 3 revealed no rupture disc failures resulting in the inoperability of BFN HPCI systems for the last three years.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.